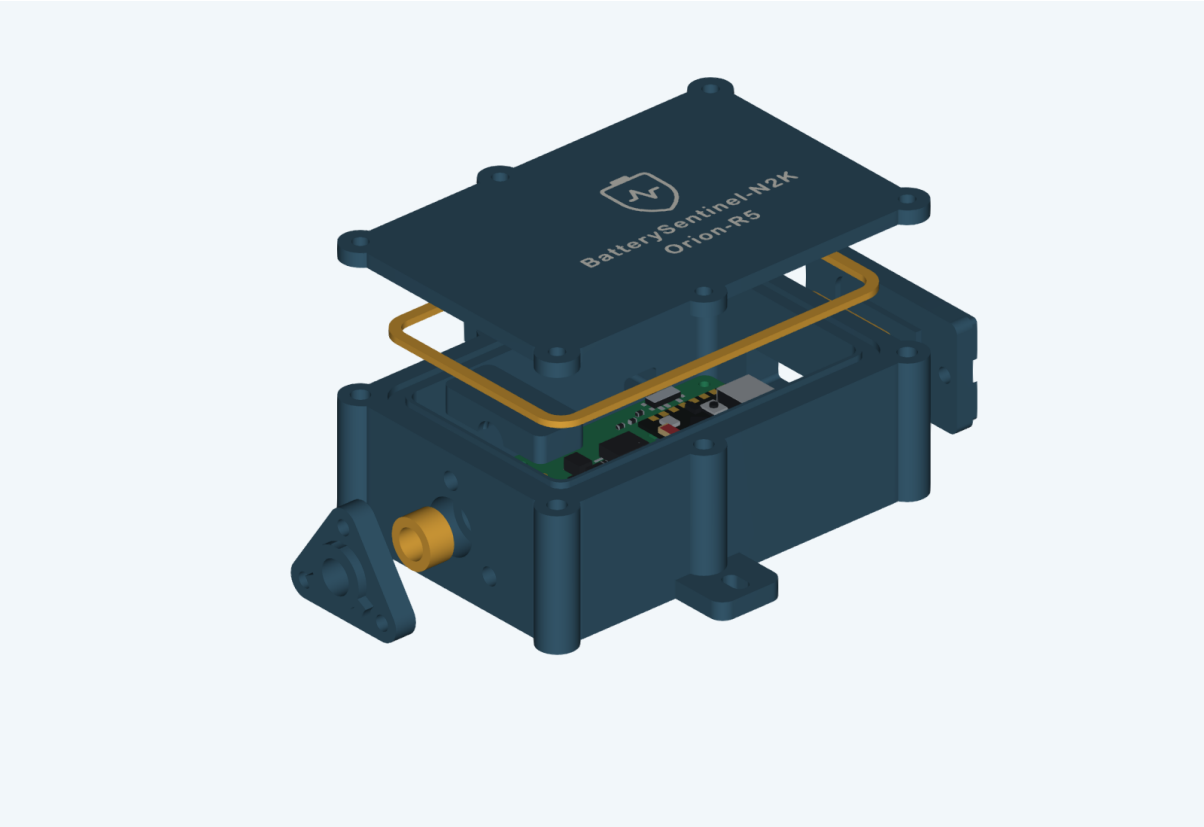
# Enclosure assembly

**V2.0 / 7 MM JACKET / SEPARATE STRAIN RELIEF**




Actual V2 CAD. Amber: seals. Screws omitted; seated seal envelopes are illustrative, not a deformation simulation.

This manual applies only to the **sealed-v2** parts. The 92 x 60 x 33.2 mm body includes a new cable bay, independent jacket clamp, three-screw pressure cap and raised face seals. External ears, service covers and fastener heads extend beyond the body dimensions.

**Prototype acceptance**

The CAD and print checks are recorded in validation.json and package-validation.json. Physical fit, insert retention, clamp force, creep and water sealing remain to be qualified. There is no tested IP rating.

Before assembly, follow the separate **Manufacturing and TPU Air calibration** manual. Use coupons before a complete case. Keep the previous printed parts separate: V2 face seals do not fit the old 1.0 mm grooves.

# Parts and tools

01 / SELECT ONE COMPLETE REVISION

| Part / material | For one enclosure |
|---|---|
| base / PETG | 1; includes PCB posts, lower saddle and clamp bosses |
| lid_body + lid_markings / PETG | 1 co-printed two-colour lid; keep the shared coordinates |
| usb_cover, service_switch_cover / PETG | 1 each; raised exterior reinforcement frame |
| cable_pressure_cap / PETG | 1; three M3 holes; flat annular contact face |
| strain_relief_upper / PETG | 1; upper ribbed jacket half-shell |
| clamp_shims_0p2 or 0p4 / PETG | One matched pair; alternatives, not both sets |
| lid_gasket, usb_gasket, service_switch_gasket / TPU Air | 1 each; continuous rounded bead |
| cable_seal_b6p8_L6p8 / TPU Air | 1 initial choice; nine variants supplied for coupon selection |

Calibration items are not enclosure components: entry_fit_coupon, face_stop_coupon_1p4 / 1p6, face_test_gasket, flow_wall_coupon and shore_comparison_pad. The sleeve filenames describe the **smallest bead bore** and **free axial length** in millimetres.

Tools: caliper, suitable insert installation tool, M2/M3 drivers, flat washers, magnifier, flat reference surface, dry absorbent paper, cable marker and controlled pull-test fixture. A Shore A durometer is needed to measure the stated hardness targets; finger feel is not a measurement.

# Fasteners and insert trials

02 / ALL EXTERIOR INSERT PILOTS STAY BLIND

| Location | Hardware / quantity | Nominal engagement |
|---|---|---|
| Lid | 6 x M3 x 8 + 0.5 mm flat washer | 4.3 mm into the pilot |
| USB + switch covers | 4 x M3 x 10 + 0.5 mm flat washer | 5.5 mm; washer seat is 4.0 mm thick |
| Cable pressure cap | 3 x M3 x 10 + 0.5 mm flat washer | 4.5 mm; washer seat is 5.0 mm thick |
| Internal clamp | 2 x M3 x 10 + 0.5 mm flat washer | 3.3 mm with 0.2 mm shim |
| PCB | 4 x M2 x 5; head <= 4.0 mm | 3.4 mm through 1.6 mm board |
| Heat-set inserts | 15 x M3; 4 x M2 | Nominal body lengths 5.7 / 4.0 mm |
| Boat mounting | 2 suitable screws + washers | Choose for mounting surface; 3.6 mm slots |

M3 pilots: nominal diameter 4.0 x depth 6.7 mm. M2 pilots: 3.2 x 4.6 mm. These pilot sizes are provisional for the purchased insert bodies; thread designation alone does not establish compatibility. Use extra inserts for trials. The cap needs one more M3 set than before; the clamp needs two more.

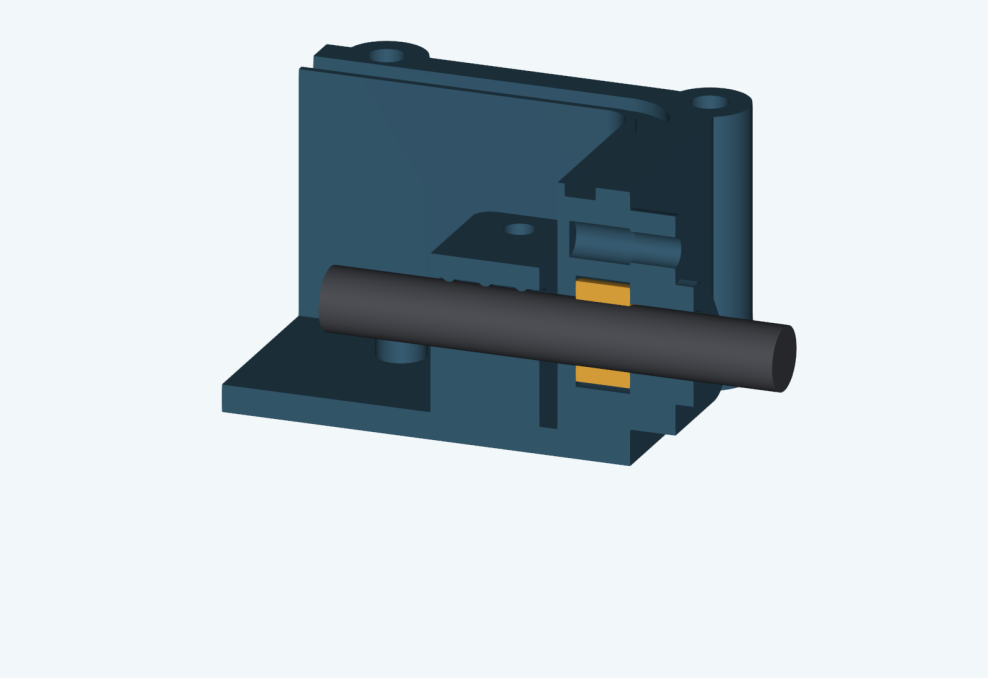
**Check before heating**

Measure the delivered inserts and screws. Validate insertion and resistance to tightening on the entry coupon. Install inserts flush and square, then allow them to cool. Do not drill a blind pilot through or bottom a screw in it. Use flat, not countersunk, heads. No tightening torque is qualified.

The engagement figures assume the stated washer thickness and fully seated inserts. Check actual lengths. A longer screw is not a remedy for a loose insert. Fasteners must suit the intended environment.

# Cable seal installation

03 / THREAD THE BARE END BEFORE SOLDERING



CAD cutaway of the cable bay. The unsplit sleeve seals; the rigid clamp behind it takes cable load. Contact envelopes simplify TPU deformation.

1 **Disconnect all power**

Unplug NMEA power and USB. SERVICE alone does not disconnect all power. Keep the PCB out for wet testing.

2 **Thread the cable**

From outside to inside: pressure cap, unsplit TPU sleeve, enclosure wall. Feed the bare-wire end through; the connector does not pass through the small hole. Do not cut a slit into the sleeve.

3 **Seat the shoulder bead**

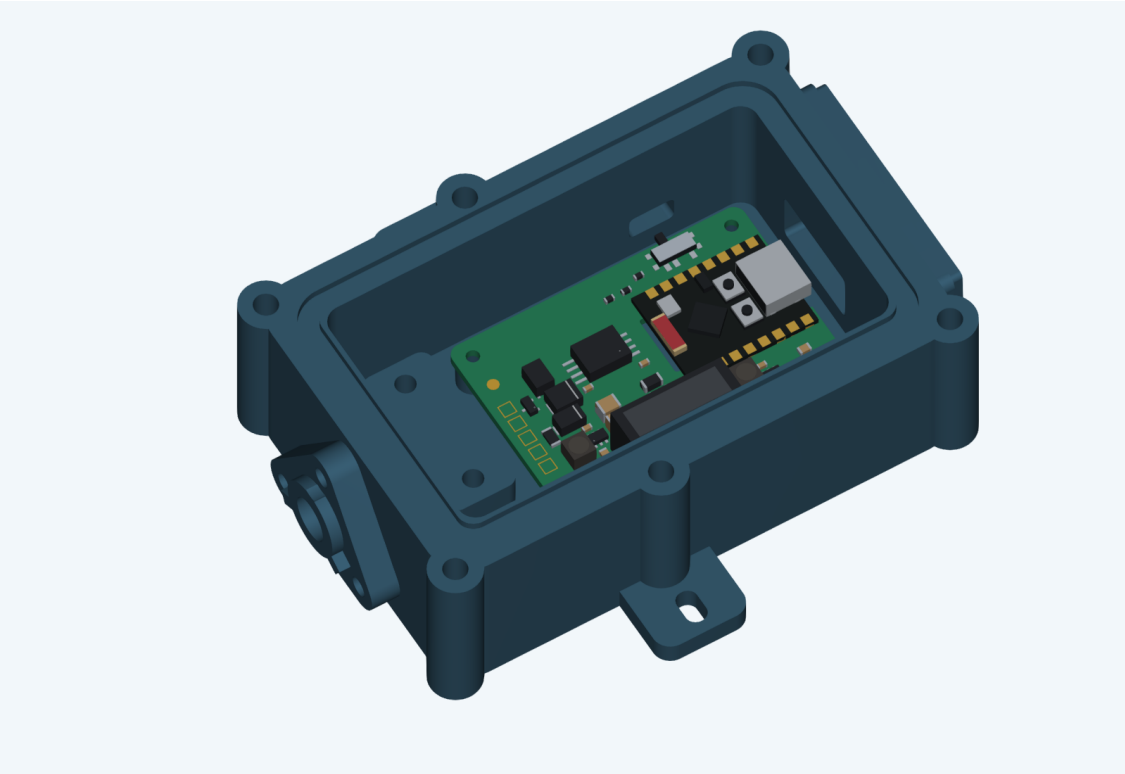
The sleeve end with the raised annular face bead points into the enclosure pocket, toward its internal shoulder. Keep the jacket intact through both the sleeve and the internal clamp.

4 **Close the three-screw cap**

Install the three screws and washers. Tighten in small alternating steps until the flat rigid lands meet the wall uniformly. Stop if the cap bows or an insert moves. Retention is adjusted at the separate clamp, not by overtightening the seal.

# Internal strain relief

04 / CLAMP ONLY THE INTACT JACKET



Current PCB/component bodies and cable bay from CAD. Check the real solder tails, wire routing and driver access during the dry fit.

Place the jacket in the lower saddle. Fit a matched shim under each upper-clamp screw ear and position the ribbed upper half-shell. Start with the **0.4 mm** pair, then compare 0.2 mm and finally no shim only if retention is insufficient. Never use unequal shims.

Bring both screws down evenly. The bosses/shims limit travel; the shell must not bend. Smaller shims increase jacket compression. Choose the largest shim that prevents movement at the required service load without damaging the jacket. The bare nominal rib throat is approximately 6.7 mm; this is an adjustable trial geometry, not a universal cable clamp setting.

Mark the jacket at both sides of the clamp. On the coupon, test clamp retention independently with seal friction removed. Apply a controlled load representative of the installation and inspect movement, jacket damage and insert movement. Never pull on individual conductors or solder joints. Leave wire slack after the clamp and support the external cable near the enclosure.

**If adjustment cannot give safe retention**

Stop and revise the saddle or add a properly sized dense liner. Do not tighten beyond the rigid stops or crush the jacket. Recheck retention after 24-48 hours and after representative temperature exposure.

# Face seals and closure

05 / BEAD ORIENTATION AND CONTROLLED COMPRESSION

| Seal | Orientation | Closed geometry |
|---|---|---|
| Lid | Flat carrier in base groove; rounded bead faces lid | 1.8 mm free -> 1.4 mm closed |
| USB cover | Flat carrier in cover groove; rounded bead faces case wall | 1.8 mm free -> 1.4 mm closed |
| Switch cover | As USB cover | 1.8 mm free -> 1.4 mm closed |
| Cable sleeve L6.8 | Raised end-face bead toward internal shoulder | 6.8 mm free -> 6.0 mm axial space |

Remove strings, support debris and raised seams without nicking the sealing faces. Reject distorted covers or interrupted beads. Seat each loop continuously, without stretch, twist or adhesive lumps. Do not double-stack seals.

1 **Close service covers**

Fit each cover with two M3 screws and washers. Alternate small turns until its hard lands meet. The seal stays inside the screw holes; do not trap wires under a cover.

2 **Close the lid**

Fit six M3 screws loosely, then tighten alternating opposite locations in several passes until the flange lands meet. Keep wire slack entirely inside the seal loop. Stop if any part bows.

**Do not mix V1 and V2 seals**

V2 face grooves are 1.4 mm deep, with nominal 22.2% total-height squeeze. The old 1.2 mm flat seals and new 1.8 mm bead seals are not interchangeable. The 1.6 mm coupon is a comparison fixture, not the production groove depth.

For service, open only in a dry location. Select SERVICE before USB connection; remove USB before RUN. Do not infer switch direction from the enclosure. Check the real USB plug can seat without forcing it, then reinstall both covers.

# Empty-case acceptance

06 / BEFORE INSTALLING ELECTRONICS

1. **Dry fit**

   Check the actual PCB, modules, plug, screws and wire routing. Verify each insert pilot remains closed and each cap reaches its hard stops. Remove the electronics again before wet tests.

2. **Exposure test**

   Put dry absorbent paper inside, away from seals, and close the empty case with its real cable. Apply the intended splash exposure. No unverified immersion depth, pressure or IP rating is assumed.

3. **Inspect**

   Dry the exterior before opening. Inspect the paper, cable pocket shoulder, lid, service covers, insert areas and FDM walls. Any moisture is a failure. Locate the leak rather than merely increasing screw torque.

4. **Repeat under use conditions**

   After cable flexing, controlled pulling, 24-48 hours of compression and opening/closing cycles, repeat the leak test. Inspect jacket and clamp marks. Record exposure, load, duration and ambient temperature.

5. **Final electrical assembly**

   Dry all parts completely. Terminate the cable according to the current Orion Bridge wiring guide, confirm continuity/polarity, mount the board with four M2 screws and retain conductor slack.

6. **Commission closed**

   After successful mechanical acceptance, check power, temperature, Wi-Fi/Bluetooth and NMEA/Garmin operation with the final closed enclosure at the intended installation location.

# Acceptance record

07 / RETAIN WITH THE ASSEMBLED DEVICE

| Record | Result / date / initials |
|---|---|
| Cable OD / ovality; sleeve filename | _____ |
| Insert body dimensions; screw lengths | _____ |
| Lid / service / cable measured Shore A | _____ |
| Filament batch; profile names; drying | _____ |
| Clamp shim; applied pull load / duration | _____ |
| Jacket movement / damage / insert movement | _____ |
| 24-48 h clamp and compression recheck | _____ |
| Initial empty-case exposure / result | _____ |
| Leak check after flexing and service cycles | _____ |
| Electrical / thermal / RF acceptance | _____ |

If foam seeps or fails recovery, use a qualified conventional seal/gland and revise its seat. Do not release a failed printed seal on the basis of Shore hardness alone.

Source of geometry: parameters.json, mechanical-contract.json, Orion-Enclosure-V2.FCStd and validation.json in sealed-v2. Dimensions are nominal CAD dimensions; physical tolerances and load ratings are not established by the digital model.